

Chapter 1

From Empirical Percentiles to Hypothesis Testing: Validating the Selection of Least-Contaminated Reference Sites

1.1 Introduction

Sediment contamination poses chronic threats to benthic communities in aquatic ecosystems, and the assessment of chemical stress requires careful selection of ecologically relevant variables. The 16 chemical variables retained in this study were chosen by ecologists and toxicologists on the basis of their known biological significance to zoobenthic organisms and related ecological concerns. Their inclusion was justified from an ecological and toxicological standpoint, reflecting the judgement that each variable carries meaningful stress information for the communities under study.

Despite their distinct contamination roles, these 16 variables do not behave as equally distinct statistical entities. As random variables measured across sampling sites, many of them exhibit strong co-variation in concentration, reflecting shared contamination sources, co-deposition pathways, or common geochemical associations, and some show highly similar distributional patterns across sites. This statistical redundancy means

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that treating the 16 raw chemical measurements as independent lines of evidence would misrepresent the true dimensionality of the chemical stress landscape: correlated variables would carry overlapping information and receive disproportionate collective weight, since the variance of their combined contribution exceeds the sum of their individual variances once their covariance is positive. As a result, the underlying structure of contamination would be obscured rather than clarified. This raises a central methodological question for the chapter: how can we extract, from 16 ecologically important but statistically overlapping chemical variables, a smaller set of latent stressors that are statistically orthogonal to one another while faithfully preserving the original contamination stress information? The value of identifying such latent stressors is precise: because they are orthogonal by construction, any contamination stress signal attributed to one latent stressor is conditionally independent of all others. This ensures that the contamination information retained within each latent stressor refers exclusively to its own domain, without carrying redundant or confounded information from the others — a property that is essential for correctly and independently partitioning contamination stress information into its respective sources.

From these orthogonal latent stressors, we adopt a specific contamination–community response perspective to synthesise the stress information they carry into a single integrated stress level for each site. This integrated stress level provides the basis for identifying sites whose communities are least affected by chemical contamination. Here it is important to distinguish two related but non-equivalent notions. A natural site refers to a community shaped under undisturbed or minimally disturbed conditions — a theoretical reference state representing biological integrity in the absence of human stressors. A least-polluted site, by contrast, is an operational concept defined relative to the observed distribution of integrated chemical stress within the study area, with sites at the low end of that distribution taken as the least-polluted.

We argue that, in this study, the least-polluted-site concept is the safer and indeed the only defensible choice of the author’s baseline. As Davies and Jackson (2006) emphasise in developing the Biological Condition Gradient, truly natural conditions are difficult to observe directly: in extensively altered regions, practitioners tend to accept the "best of what remains" as the apparent optimum, because the original

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natural state can no longer be visualised, which truncates the perspective on which assessment is based. Moreover, they note that least-disturbed reference sites differ markedly across regions in their degree of departure from historical or natural condition. Direct observation of a genuinely natural community is therefore not attainable from the available data; the least-polluted sites, identified from the low tail of the integrated stress distribution, serve as the closest observable and comparable approximation to natural community organisation.

The ecological significance of these least-polluted sites follows directly. Within them, community structure is expected to be governed primarily by habitat and environmental factors rather than by chemical stress, so they retain the information of naturally organised communities. They thus provide an observable, comparable natural benchmark on many levels — essential both for understanding natural community composition and for characterising the community changes induced by contamination stress.

The central contribution of this chapter is methodological. Rather than delineating the least-contaminated reference group by a fixed empirical percentile—a convention widely adopted yet seldom verified—we recast the choice of threshold as a question to be tested. Using two complementary resampling-based hypothesis tests, we ask whether a candidate reference group of a given size genuinely carries weaker contamination-constrained community variation than would be expected by chance, and we use this evidence to identify the largest defensible least-contaminated set rather than fixing it at an arbitrary cut-off. In doing so, the chapter provides a statistical basis on which the selection of a least-contaminated baseline—and the percentile threshold that defines it—can be justified rather than merely assumed.

1.2 Methods

1.2.1 Principle Component Analysis/R-model Ordination on Chemical Matrix

In linear algebra, the eigen-decomposition of a matrix identifies a set of characteristic directions along which a linear transformation acts purely by scaling, and within a measure space it resolves the structure of that space into mutually orthogonal axes whose orthogonality can be formally proven from the symmetry of the underlying matrix (Strang, 2006). When this decomposition is applied to a data matrix, it acquires a statistical interpretation: Principal Component Analysis is an eigen-decomposition performed after a descriptive space — conventionally the column space — has been chosen for the data matrix, and in numerical ecology it is classified among the R-mode ordination methods that operate under a Euclidean distance measure of dissimilarity (Legendre and Legendre, 2012). PCA extracts the eigenvectors from the correlation matrix of the column space, and the resulting orthogonal eigenvectors are scaled so as to preserve, as far as possible, the structure and magnitude of the data within the original descriptive space. The values of the original variables are then projected onto these orthogonal vectors and come to lie in a new descriptive space spanned by the ordination axes, the principal components. It must be emphasised that orthogonality holds only among the newly constructed principal components; the covariance among the original variables is still partly retained within these components. This is reflected in the fact that the variance of each principal component preserves not only the original quantity of information (variance) but also the covariance information among the original variables.

In this study, the structure and magnitude of the variance carried by the principal components themselves are not our central concern. What concerns us is rather the assurance that the principal components preserve the original information content together with their mutual orthogonality, that is, their statistical independence. In this way, when we discuss the contamination stress information of one or several stressors at a given site, we can ensure that it is independently and completely retained within the domain to which the latent stressor(s) refer, allowing the contamination

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variation arising from different latent stressors to be correctly distinguished.

A brief proof of the mathematical properties underlying PCA is provided in the Appendix 1.5.1). Performing PCA on the chemical matrix, then rotating and selecting the leading principal components, yields a set of latent stressors that quantify the contamination stress information. The full procedure, from the input chemical matrix to the output latent stressors, is summarised in Algorithm 1.

Algorithm 1: Construction of the PC-based chemical stressor matrix

Data: Site-by-chemical matrix $\mathbf{M}_{310 \times 16}$.

Result: PC-based chemical stressor matrix $\mathbf{S}_{310 \times p}$.

- 1 Log-transform and standardise, then form the correlation matrix:

$$\mathbf{M}_{310 \times 16} \rightarrow \mathbf{M}'_{310 \times 16} \xrightarrow{\text{correlation}} \mathbf{cor}_{16 \times 16}$$

- 2 Apply PCA, quartimax rotation, and loading selection:

$$\mathbf{cor}_{16 \times 16} \rightarrow \mathbf{Loadings}_{16 \times p'} \xrightarrow{\text{rotation}} \mathbf{Loadings}^*_{16 \times p'} \xrightarrow{\text{selection}} \mathbf{Loadings}^*_{16 \times p}$$

- 3 Project the processed chemical matrix onto the selected rotated loading

$$\text{space: } \mathbf{S}_{310 \times p} = \mathbf{M}'_{310 \times 16} \mathbf{Loadings}^*_{16 \times p}$$

- 4 **return** $\mathbf{S}_{310 \times p}$
-

1.2.2 Latent Stressors and Contamination Cumulative Impact

In multiple-stressor ecology, stressors are commonly defined externally as ecologically interpretable pressures, such as nutrient enrichment, habitat degradation, flow alteration, or pollution. Studies such as those of Halpern et al. and Birk et al. adopt this perspective to examine how multiple recognisable environmental pressures combine, dominate, or interact in shaping biological responses (Halpern et al., 2008 [3]; Birk et al., 2020 [2]).

In the present study, by contrast, the chemical stressors are not directly defined pressure categories. They are latent contamination gradients extracted through the procedure of the preceding subsection, and are best interpreted as contamination syndromes (Huang et al. [4] and Loska and Wiechula [5]). The latent stressors here therefore do not denote individual pollutants, but rather structured chemical pres-

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sures potentially experienced by the benthic communities.

From these latent stressors, an integrated contamination stress is derived for each site. We adopt the perspective that the community-level effects of the latent stressors are *additive*: their individual stress contributions are summed to yield a single integrated contamination stress. This quantity is denoted the Sum Relative Stress (SumRel), which expresses the position of a site relative to all other sites within the summed-stressor value space. Accordingly, the SumRel of site i in the 310 sites is calculated as:

$$SumRel_i = \sum_{j=1}^p s_{ij}, \quad s_{ij} \in \mathbf{S}_{310 \times p}$$

Different contamination–community response perspectives may legitimately coexist. The additive view adopted here assumes that summing independent stress values reflects the combined influence of the stressors on community composition (Halpern et al., 2008 [3]), whereas a dominance view holds instead that a single prevailing stressor obscures the others and governs the response, so that the integrated effect is set by the highest stressor value alone (Birk et al., 2020 [2]). We construct the integrated stress measure (SumRel) under the additive perspective, while acknowledging that the choice among such perspectives is one that ultimately be left to ecologists to judge.

1.2.3 Least-Contaminated Sites Selection and Validation

Proceeding from the integrated stress level, we define the *least-contaminated sites* as those lying at the low tail of the observed distribution of integrated contamination stress (SumRel). These sites constitute the least-contaminated group, whose taxa communities are expected to be the least affected by contamination and thus to serve as a benchmark, or baseline.

Determining this group generally requires a threshold to separate the least-contaminated sites from the remainder. In the study of Zhang (Zhang, 2008 [8]), this threshold was set at the 20th percentile of integrated stress. Such a 20% cut-off is, in most cases, chosen empirically (Australian and New Zealand Governments, 2018 [1]; Ministry for

the Environment [6]). While this approach does capture the sites at the low end of the observed stress distribution, it neither examines how a fixed-percentile choice (20%) weakens the least-contamination assumptions underlying the delineated group size when applied across differing contamination settings (Stoddard et al., 2006 [7]), nor verifies that a low-percentile group of a given size exerts a significantly weaker influence on community composition than other groups of the same size.

Prescribing an absolute count threshold (for instance, 50 or 60 sites) applicable across all contamination settings is likewise neither attainable nor interpretable, since such a number depends on the distribution of stress levels in the study context and on the researcher’s judgement of what constitutes, and how to constitute, an ecologically “low” level of contamination.

1.2.4 A resampling-based hypothesis testing design for validating least-contaminated effects on community of least-contaminated sites

In this section we continue to adopt the 20th percentile to select the least-contaminated sites, but design a hypothesis test, based on a multivariate-regression resampling procedure, to verify whether the level of contamination impact on the community within the least-contaminated group is significantly lower than that of other, randomly drawn groups. The method draws on the first half of Redundancy Analysis (RDA), measuring the magnitude of contamination impact on the community within each group by fitting a multivariate regression model.

Specifically, for a given set of m sites with retained latent-stressor matrix $\mathbf{X}$ and community composition matrix $\mathbf{Y}$, we regress each species record $\mathbf{y}$ in $\mathbf{Y}$ on $\mathbf{X}$. The fitted records of all species are collected into $\hat{\mathbf{Y}}$, a matrix of the same dimensions as $\mathbf{Y}$. The variation in $\hat{\mathbf{Y}}$ is constrained by $\mathbf{X}$ and is taken as a measure of the community variation induced by the latent stressors that $\mathbf{X}$ represents. The adjusted coefficient of determination $\bar{R}^2$ of the model, that is, the adjusted ratio $\frac{\text{variation of } \hat{\mathbf{Y}}}{\text{variation of } \mathbf{Y}}$, expresses the proportion of total community variation that is accounted for under the constraint of the stressors, in other words the magnitude of contamination impact

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on the community. The exact computation of $\bar{R}^2$ from the regression fit is given in Appendix 1.5.2.

In this test, we randomly draw B site sets, each of size m_{ref} , denoted $\mathbf{S}_b$, $b \in \{1, \dots, B\}$. The above multivariate regression model is fitted on each set, yielding a distribution of B values $\bar{R}_b^2$ that reflect the magnitude of contamination impact on the community under random sampling. The set of least-contaminated sites comprising the lowest 20th percentile is denoted $\mathbf{S}_{\text{ref}}$; fitting the same model on this set yields a single value $\bar{R}_{\text{ref}}^2$, reflecting the magnitude of contamination impact on the community within the least-contaminated set.

We then compare $\bar{R}_{\text{ref}}^2$ against the B values $\bar{R}_b^2$ obtained by random resampling. If $\bar{R}_{\text{ref}}^2$ is significantly lower than the randomly distributed $\bar{R}_b^2$ values, the level of contamination impact on the community within the least-contaminated set is taken to be significantly lower than that observed under random sampling, thereby validating that contamination stress exerts a lower level of influence on community composition within the least-contaminated set.

The specific computational procedure is summarised in Algorithm 2.

Algorithm 2: Between-group resampling test for reduced contamination impact in least-contaminated sites

Data: Latent-stressor matrix $\mathbf{X}$, community matrix $\mathbf{Y}$, reference size m_{ref} , replicates B .

Result: Significance of reduced contamination impact in $\mathbf{S}_{\text{ref}}$.

- 1 Fit the reference set: take $\mathbf{S}_{\text{ref}}$ as the lowest 20% of sites by SumRel, regress $\mathbf{Y}$ on $\mathbf{X}$, and record $\bar{R}_{\text{ref}}^2$
 - 2 **for** $b \leftarrow 1$ **to** B **do**
 - 3 Draw a random site set $\mathbf{S}_b$ of size m_{ref} , fit the same regression, and record $\bar{R}_b^2$
 - 4 Compare $\bar{R}_{\text{ref}}^2$ with the null distribution $\{\bar{R}_b^2\}_{b=1}^B$
 - 5 **return** $\mathbf{S}_{\text{ref}}$ is validated if $\bar{R}_{\text{ref}}^2$ lies significantly below $\{\bar{R}_b^2\}$
-

This comparison constitutes a nonparametric, resampling-based hypothesis test. Rather than referring $\bar{R}_{\text{ref}}^2$ to a theoretical sampling distribution, we construct an empirical null distribution $\{\bar{R}_b^2\}_{b=1}^B$ directly by random resampling, so that no parametric as-

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sumptions on the distribution of the test statistic are required. The hypotheses are stated in Table 1.1. Under H_0 , the least-contaminated set behaves no differently from an arbitrary set of the same size, and $\bar{R}_{\text{ref}}^2$ is an ordinary draw from the null distribution; under the one-sided alternative H_A , the least-contaminated set exhibits weaker stressor-constrained variation, so that $\bar{R}_{\text{ref}}^2$ falls in the lower tail of $\{\bar{R}_b^2\}$. The corresponding left-tailed empirical p -value is

$$p = \frac{1 + \#\{b : \bar{R}_b^2 \leq \bar{R}_{\text{ref}}^2\}}{B + 1},$$

and H_0 is rejected at level α when $p \leq \alpha$, validating that contamination stress exerts a significantly weaker influence on community composition within the least-contaminated set than under random sampling.

Table 1.1: Hypotheses for the resampling-based test of reduced stressor-constrained variation in the least-contaminated set.

H_i	Statement
H_0	The m_{ref} lowest-scored sites do not show weaker stressor-constrained variation than a random m_{ref} -site subset.
H_A	The m_{ref} lowest-scored sites show weaker stressor-constrained variation than a random m_{ref} -site subset.

The between-group test above asks whether the least-contaminated set stands out *relative to other sets of equal size*. A complementary question can be posed about a single set in isolation: judged only against its own data, does the set carry a real stressor–community association, or none at all? We address this with a within-group permutation test.

For an observed set $\mathbf{S}_b$, we randomly permute the rows of its community matrix $\mathbf{Y}_b$ a total of B' times. Each permutation breaks the site-wise correspondence between stressors and community records, so the refitted value $\bar{R}_b^{2,\text{perm}}$ measures the stressor-constrained variation attainable by chance alone. The B' permutations form a null distribution against which the observed $\bar{R}_b^2$ is judged.

For the least-contaminated set, we are interested in the opposite outcome to a con-

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ventional significance test: the set qualifies as a natural benchmark precisely when its stressor–community association cannot be distinguished from chance. If the observed $\bar{R}_b^2$ does *not* lie in the upper tail of the permutation distribution—that is, if it is no more extreme than the constrained variation produced by random row reassignment—then the contamination gradient exerts no detectable constraint on community composition within the set. The corresponding right-tailed permutation p -value,

$$p_b = \frac{1 + \#\{j : \bar{R}_{b,j}^{2,\text{perm}} \geq \bar{R}_b^2\}}{B' + 1},$$

is the probability of obtaining an association at least as strong as the observed one when stressors and community are in truth unrelated. A *large* p_b therefore indicates that the observed association is indistinguishable from random reassignment, so that the set carries essentially no contamination-constrained structure. Ecologically, failing to reject the null hypothesis is the desired result: it characterises a community organised independently of the contamination gradient—the state expected of a least-contaminated assemblage governed by natural environmental factors—whereas a small p_b would instead reveal a set within which contamination measurably shapes community composition. The hypotheses tested here can be expressed in the form of Table 1.2, and the detailed computation can be summarised in the form of Algorithm 3.

Table 1.2: Hypotheses for the within-group permutation test of stressor-constrained variation. A least-contaminated set is expected to *fail to reject* H_0 , indicating no detectable contamination constraint on community composition.

H_i	Statement
H_0	Within $\mathbf{S}_b$, the stressors constrain community variation no more strongly than under random row reassignment of $\mathbf{Y}_b$.
H_A	Within $\mathbf{S}_b$, the stressors constrain community variation significantly more strongly than under random row reassignment of $\mathbf{Y}_b$.

Algorithm 3: Within-group permutation test of stressor–community association

Data: Site set $\mathbf{S}_b = (\mathbf{X}, \mathbf{Y}_b)$, permutations B' .

Result: Significance of the stressor–community association within $\mathbf{S}_b$.

- 1 Fit the observed set: regress $\mathbf{Y}_b$ on $\mathbf{X}$ and record $\bar{R}_b^2$
 - 2 **for** $j \leftarrow 1$ **to** B' **do**
 - 3 Permute the rows of $\mathbf{Y}_b$ to obtain $\mathbf{Y}_b^{\text{perm}}$, refit on $\mathbf{X}$, and record $\bar{R}_{b,j}^{2,\text{perm}}$
 - 4 Compute the right-tailed p -value p_b from $\bar{R}_b^2$ and $\{\bar{R}_{b,j}^{2,\text{perm}}\}_{j=1}^{B'}$
 - 5 **return** $\mathbf{S}_b$ *behaves as a natural benchmark if $p_b > \alpha$ (the contamination constraint is indistinguishable from random reassignment)*
-

Together, the between-group resampling test of Algorithm 2 and the within-group permutation test of Algorithm 3 probe the least-contaminated set from two complementary angles—its standing relative to comparable sets, and the strength of its own internal stressor–community signal. By scanning candidate sets of increasing size ordered by SumRel and tracking the size at which p_b falls below α , we locate the point beyond which contamination begins to constrain community composition—thereby identifying a defensible least-contaminated group rather than fixing it at a fixed conventional percentile.

The scanning procedure is an extension of Algorithm 2 and 3, increasing the least-contaminated group size from 15 to 310 sites in stepwise increments of 5 sites and performing the two tests at each step. At every size m_{ref} , the reference curves $\bar{R}_{\text{ref}}^2$ and p_{ref} are recorded together with the random-subset bands of the same statistics, as summarised in Algorithm 4.

Applied across cases, this procedure behaves in an informative way: in the *corridor case*, where lightly contaminated sites are comparatively abundant, it supports a relatively large least-contaminated set, so that the 20% threshold is a reasonable choice; in the *Detroit River case*, where such sites are scarcer, it favours a smaller set, so that the same 20% threshold becomes difficult to justify.

Algorithm 4: Threshold scan across least-contaminated subset sizes

Data: Latent-stressor matrix $\mathbf{X}$, community matrix $\mathbf{Y}$, SumRel-ordered
sites, step $\delta = 5$.

Result: Curves $\bar{R}_{\text{ref}}^2(m_{\text{ref}})$ and $p_{\text{ref}}(m_{\text{ref}})$ with random-subset bands.

```

1 for  $k \leftarrow 0$  to 59 do
2    $m_{\text{ref}} \leftarrow 15 + k \delta$ 
3   Take  $\mathbf{S}_{\text{ref}}$  as the  $m_{\text{ref}}$  lowest-SumRel sites, and record  $\bar{R}_{\text{ref}}^2$  and  $p_{\text{ref}}$  with
   their random-subset bands (Algorithms 2–3)
4 return  $\{\bar{R}_{\text{ref}}^2(m_{\text{ref}})\}$  and  $\{p_{\text{ref}}(m_{\text{ref}})\}$ 

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1.3 Results

The results

1.4 Tables and Figures

Table 1.3: Rotated PCA loadings and variance summary for the first six principal components

Chemicals	PC1	PC2	PC3	PC4	PC5	PC6
Cr	0.9174	0.2387	0.1500	-0.0043	-0.0663	0.0623
Zn	0.8865	-0.0024	-0.0226	-0.0688	0.1449	0.0614
Ni	0.8861	0.0193	-0.0291	0.0311	-0.2636	-0.0020
Cu	0.8839	0.2188	0.1136	-0.0860	-0.0045	-0.0379
total PCB	0.8364	-0.3031	0.0049	0.1665	0.1523	0.0210
Cd	0.7387	-0.0877	0.4223	0.0595	0.1766	0.1071
Co	0.6772	0.6583	0.0279	0.0688	0.1882	0.0637
p,p'-DDE	0.6385	-0.1622	-0.1223	0.4012	0.2385	0.0789
Mn	0.5012	0.3860	0.3665	0.1539	-0.3701	0.1585
Ca	0.2516	0.1462	0.0287	0.1107	0.0568	0.0954
Al	0.4609	0.8085	0.0437	0.0214	0.1118	0.1404
Pb	0.3790	-0.0207	0.8037	0.0162	-0.3245	-0.0084
As	0.3145	0.1097	0.7899	-0.0134	0.3535	0.0510
OCS	0.1816	0.0541	0.0238	0.9356	0.0096	0.0095
Hg	0.4581	0.2655	0.0210	0.0609	0.7510	0.0320
Fe	0.4722	0.1712	0.0579	0.0212	0.0190	0.8420
Explained Var.	6.5407	1.6036	1.6422	1.1264	1.1890	0.7980
Prop. Var.	0.4088	0.1002	0.1026	0.0704	0.0743	0.0499
Cum. Prop.	0.4088	0.5090	0.6117	0.6821	0.7564	0.8063

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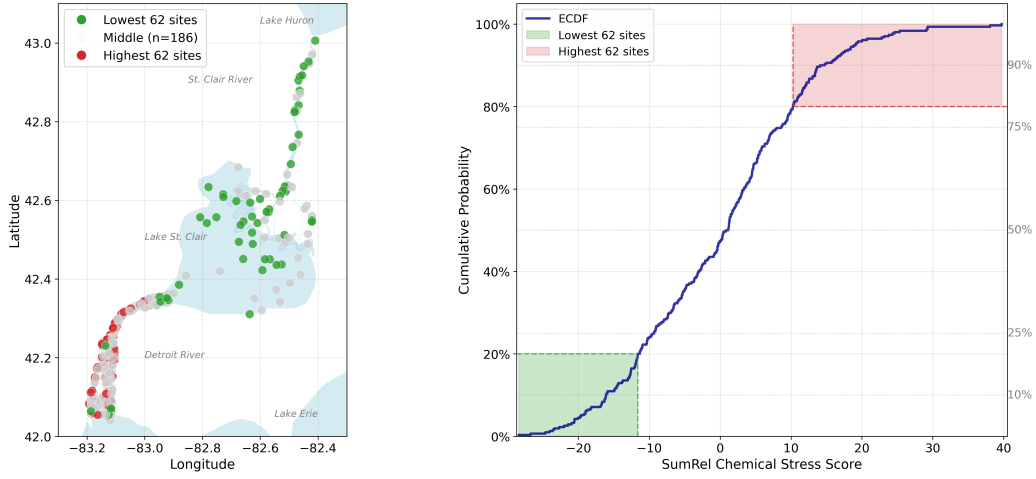


Figure 1.1: Sum Relative Contamination Score (SumRel) across the 310 corridor sites. Left: corridor map of the lowest 20% (62 sites, green), the highest 62 sites (red), and the middle 186 sites (grey). Right: SumRel ECDF, with the lowest and highest 62 sites bifurcated at the 20th and 80th percentiles.

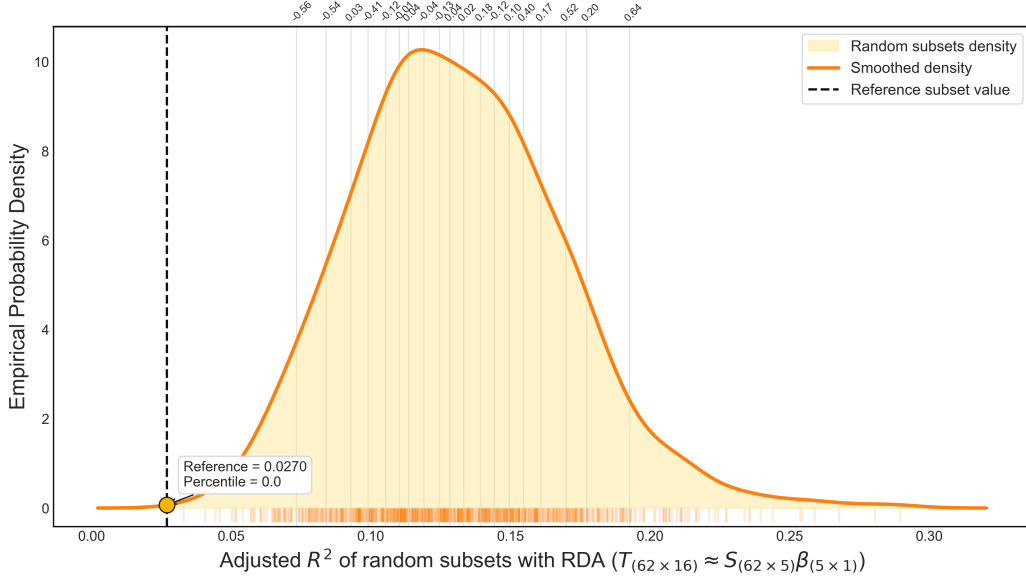


Figure 1.2: Between-group resampling test (Algorithm 2) at $m_{\text{ref}} = 62$: empirical $\bar{R}^2$ distribution of 1000 random 62-site subsets. The top axis shows average contamination scores of the 50 nearest subsets per location, with tick density proportional to probability mass. The black line marks the reference $\bar{R}^2$ (percentile = 0.0); with $p \leq 0.05$, H_0 is rejected—the least-contaminated set is significantly cleaner than random sets.

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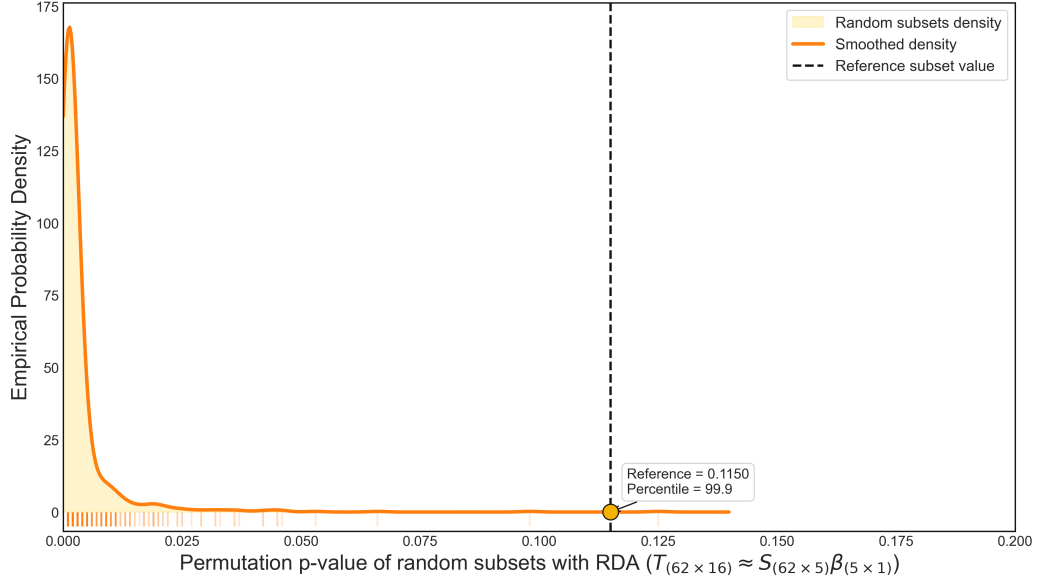


Figure 1.3: Within-group permutation test (Algorithm 3) at $m_{\text{ref}} = 62$: empirical distribution of permutation p -values from 1000 resampled subsets, each permuted 1000 times. The black line marks the reference p -value (percentile = 99.9); with $p \geq 0.05$, H_0 is not rejected—within the least-contaminated set the constraint is indistinguishable from chance, as expected of a natural benchmark.

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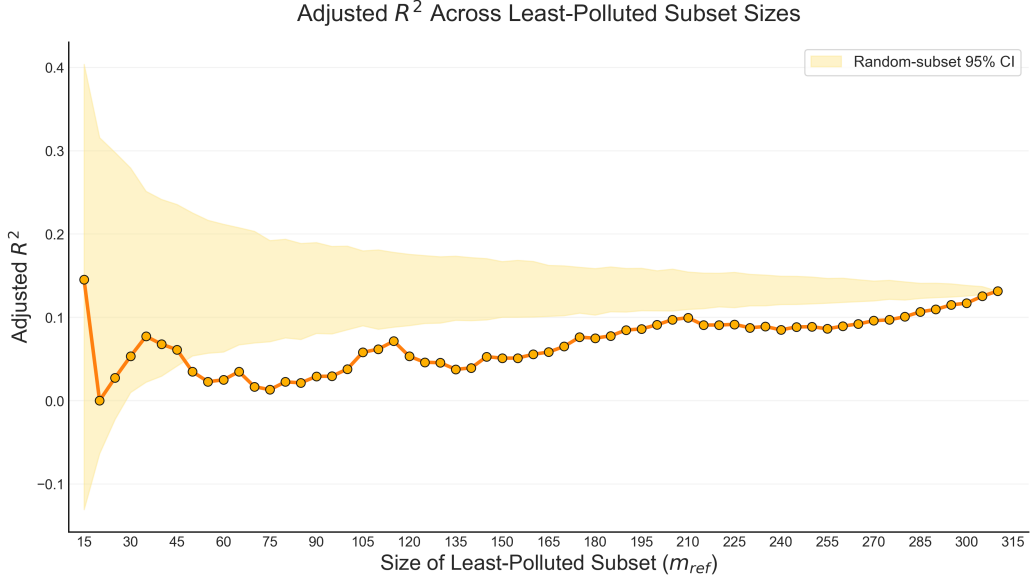


Figure 1.4: Threshold scan of $\bar{R}^2$ across subset sizes (Algorithm 4). The orange curve traces $\bar{R}^2_{\text{ref}}$ of the m_{ref} lowest-SumRel sites for $m_{\text{ref}} = 15$ to 310; the band is the random-subset 95% interval. The curve hugs the lower edge at small sizes and rises toward the band centre as more contaminated sites are admitted.

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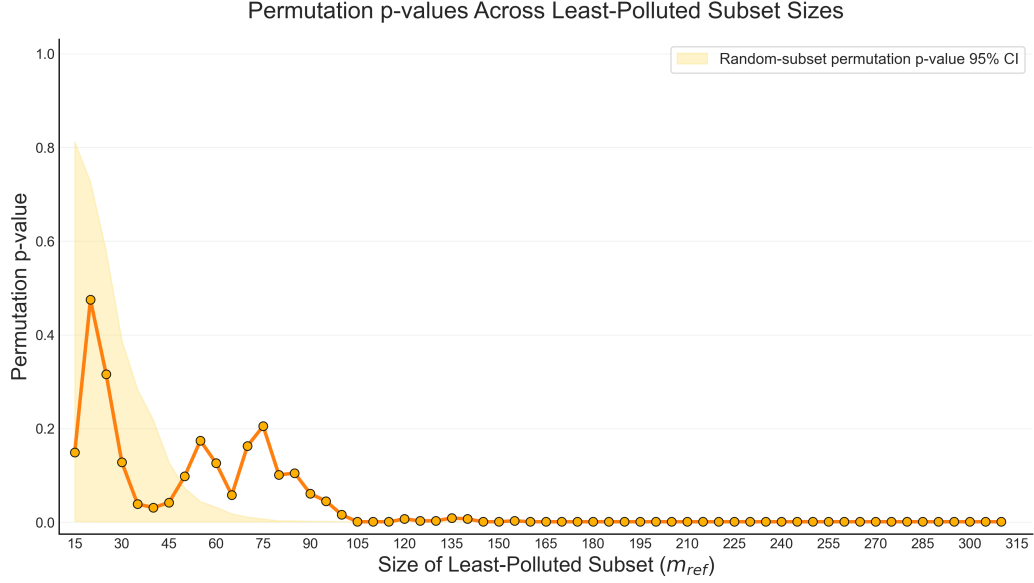


Figure 1.5: Threshold scan of the permutation p -value across subset sizes (Algorithm 4). The orange curve traces p_{ref} of the m_{ref} lowest-SumRel sites for $m_{\text{ref}} = 15$ to 310; the band is the random-subset 95% interval. The reference p_{ref} stays high for small sets and collapses toward zero beyond $m_{\text{ref}} \approx 100$, marking where contamination begins to constrain community composition.

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1.5 Appendix

1.5.1 Orthogonality and variance preservation of principal components

Setup. Let $\mathbf{X}$ be an $n \times p$ data matrix recording p descriptors (variables) over n objects (sites). Assume each column has been centred and standardised to unit variance, so that the $p \times p$ correlation matrix of the column space is

$$\mathbf{R} = \frac{1}{n-1} \mathbf{X}^\top \mathbf{X}.$$

By construction $\mathbf{R}$ is *symmetric* ($\mathbf{R}^\top = \mathbf{R}$) and *positive semi-definite*, since for any vector $\mathbf{u} \in \mathbb{R}^p$,

$$\mathbf{u}^\top \mathbf{R} \mathbf{u} = \frac{1}{n-1} (\mathbf{X}\mathbf{u})^\top (\mathbf{X}\mathbf{u}) = \frac{1}{n-1} \|\mathbf{X}\mathbf{u}\|^2 \geq 0.$$

Step 1 — Eigen-decomposition exists and yields an orthonormal basis. By the spectral theorem, every real symmetric matrix admits a decomposition

$$\mathbf{R} \mathbf{v}_k = \lambda_k \mathbf{v}_k, \quad k = 1, \dots, p,$$

with *real* eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$ (non-negative by positive semi-definiteness). The eigenvectors are mutually orthogonal: for $\lambda_i \neq \lambda_j$,

$$\lambda_i \mathbf{v}_i^\top \mathbf{v}_j = (\mathbf{R}\mathbf{v}_i)^\top \mathbf{v}_j = \mathbf{v}_i^\top \mathbf{R}\mathbf{v}_j = \lambda_j \mathbf{v}_i^\top \mathbf{v}_j \implies (\lambda_i - \lambda_j) \mathbf{v}_i^\top \mathbf{v}_j = 0 \implies \mathbf{v}_i^\top \mathbf{v}_j = 0.$$

Hence we may normalise them into an orthonormal set, collected as columns of an

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orthogonal matrix $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_p]$ with $\mathbf{V}^\top \mathbf{V} = \mathbf{V} \mathbf{V}^\top = \mathbf{I}$, giving

$$\mathbf{R} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top, \quad \mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_p).$$

Step 2 — The eigenvectors preserve the structure of $\mathbf{R}$. Because $\mathbf{V}$ is orthogonal, the decomposition is an *exact* change of basis: no information in $\mathbf{R}$ is lost, only re-expressed. The eigenvectors form the directions along which $\mathbf{R}$ acts purely by scaling, and the total variance is conserved,

$$\text{tr}(\mathbf{R}) = \text{tr}(\mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top) = \text{tr}(\mathbf{\Lambda}) = \sum_{k=1}^p \lambda_k = p,$$

so each λ_k is precisely the amount of total variance carried by direction $\mathbf{v}_k$. The structure of $\mathbf{R}$ is thus repackaged, without distortion, into ordered orthogonal axes.

Step 3 — Projection gives orthogonal scores that retain maximal variation.

Project the descriptors onto the eigenvectors to obtain the principal-component scores

$$\mathbf{F} = \mathbf{X} \mathbf{V}, \quad \mathbf{f}_k = \mathbf{X} \mathbf{v}_k.$$

These new axes are *mutually uncorrelated*: their covariance matrix is diagonal,

$$\frac{1}{n-1} \mathbf{F}^\top \mathbf{F} = \mathbf{V}^\top \left(\frac{1}{n-1} \mathbf{X}^\top \mathbf{X} \right) \mathbf{V} = \mathbf{V}^\top \mathbf{R} \mathbf{V} = \mathbf{\Lambda},$$

so $\text{Var}(\mathbf{f}_k) = \lambda_k$ and $\text{Cov}(\mathbf{f}_i, \mathbf{f}_j) = 0$ for $i \neq j$. Because the λ_k are ordered, the first few components carry the largest share of variation: retaining the leading $m < p$ axes captures the fraction

$$\frac{\sum_{k=1}^m \lambda_k}{\sum_{k=1}^p \lambda_k}$$

of the total variance, the maximum attainable by any m orthogonal directions (Rayleigh–Ritz). The optionally *scaled* eigenvectors $\mathbf{v}_k \sqrt{\lambda_k}$ (distance/correlation biplot scaling) rescale axis length to reflect λ_k while leaving orthogonality untouched.

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1.5.2 Computation of the adjusted constrained variation $\bar{R}^2$

Setup. Consider a site set of m objects with latent-stressor matrix $\mathbf{X} \in \mathbb{R}^{m \times p}$ (an intercept column prepended) and Hellinger-transformed, column-centred community matrix $\mathbf{Y} \in \mathbb{R}^{m \times q}$ over q taxa. The aim is to measure how much community variation the stressors constrain.

Step 1 — Multivariate least-squares fit. Regressing all taxon columns of $\mathbf{Y}$ on $\mathbf{X}$ gives the fitted values

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}, \quad \mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top,$$

where $\mathbf{H}$ projects onto the column space of $\mathbf{X}$; $\hat{\mathbf{Y}}$ is thus the portion of the community response lying within the stressor space.

Step 2 — Variation partitioning. With $\mathbf{H}$ symmetric and idempotent, the explained fraction of community variation is

$$R^2 = \frac{\text{SS}_{\text{fit}}}{\text{SS}_{\text{tot}}} = \frac{\text{tr}(\mathbf{Y}^\top \mathbf{H} \mathbf{Y})}{\text{tr}(\mathbf{Y}^\top \mathbf{Y})}.$$

Step 3 — Adjustment for dimensionality. Since R^2 rises with p and falls with m , sets of differing size are not directly comparable; Ezekiel's degrees-of-freedom correction gives

$$\bar{R}^2 = 1 - (1 - R^2) \frac{m - 1}{m - p - 1}.$$

This $\bar{R}^2$ is the statistic compared across reference, random, and permuted sets of common size m_{ref} , so that differences reflect the strength of the stressor–community constraint rather than the dimensions of the fit.